

Pythagoras

Proofs you can see

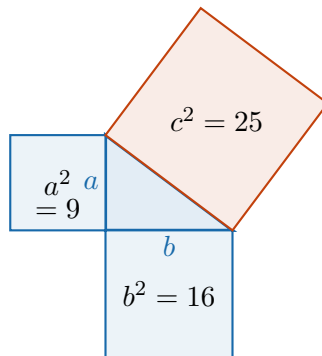
The setup. A *right triangle* has one 90-degree corner. The two short sides touching it are the *legs*, a and b . The long side across from it is the *hypotenuse*, c . The claim:

$$a^2 + b^2 = c^2.$$

We are not going to memorize it. We are going to *see* why it must be true.

Build 1: three squares on a 3-4-5 triangle

On grid paper, draw a right triangle with legs 3 and 4. On each side, draw the square that sticks out away from the triangle (like the picture). Count the boxes in each square.



Add the two leg squares: $9 + 16 = \underline{\hspace{2cm}}$. Now look at the big tilted square. What is its area? $\underline{\hspace{2cm}}$

Draw your own version on grid paper here:

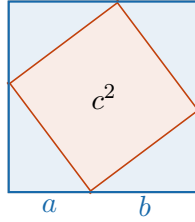
Build 2: count a tilted square

You cannot read off the side of a tilted square, so use this trick. Box the tilted square inside the smallest *upright* square (for 3-4-5 it is a 7×7 box, area 49). Four copies of your triangle sit in the corners, each of area $\frac{1}{2}(3)(4) = 6$.

Tilted square area = $49 - 4 \times 6 = \underline{\hspace{2cm}}$. Does it match Build 1? $\underline{\hspace{2cm}}$

See the proof: the tilted square

A big square of side $a + b$ holds four copies of the triangle, leaving a tilted square of side c in the middle.



Find the big square's area two ways, then set them equal.

- As one square: $(a + b)^2 =$ _____
- As pieces: four triangles + the hole = _____

Set them equal, cancel the $2ab$, and write what is left:

Push further

1. **The triple machine.** Pick whole numbers $m > n$. Then $(m^2 - n^2, 2mn, m^2 + n^2)$ is always a Pythagorean triple. Fill the table and check each one squares correctly.

m	n	$m^2 - n^2$	$2mn$	$m^2 + n^2$	check
2	1				
3	2				
4	1				
4	3				

2. **Competition gem (the converse).** The theorem says: right angle $\Rightarrow a^2 + b^2 = c^2$. The converse runs the arrow the other way: if $a^2 + b^2 = c^2$, the angle opposite c *must* be a right angle. This is why a knotted 3-4-5 rope makes a perfect right corner. Prove it. (Hint: build a brand-new right triangle with legs a and b . What is its hypotenuse, and how does that pin down your triangle's angle?)

3. **True or false?** Guess first, then ask.

- $a^2 + b^2 = c^2$ works for *any* triangle, not just right ones. T / F
- A box with edges a, b, c has space diagonal d with $d^2 = a^2 + b^2 + c^2$. T / F
- There are only finitely many Pythagorean triples. T / F